Safety and Rollback Mechanisms for Objective Misinterpretation in Agentic AI Systems

Research Paper

Abstract

Agentic artificial intelligence (AI) systems are increasingly capable of planning, reasoning, using external tools, and executing actions with limited human intervention. While these capabilities can improve efficiency and autonomy, they also introduce a significant safety concern: an agent may interpret its assigned objective differently from what its developer or user intended. This problem, commonly associated with objective misalignment, specification errors, reward misspecification, and goal misgeneralization, can cause an otherwise capable system to take harmful or undesirable actions. Traditional AI safety techniques primarily focus on preventing unsafe outputs, but agentic systems require additional mechanisms capable of detecting incorrect objectives during execution and safely reversing their consequences. This paper examines objective misinterpretation in agentic AI systems and proposes a safety architecture combining objective validation, runtime monitoring, action constraints, human approval, transaction-style execution, checkpoints, and rollback mechanisms. The paper also discusses the limitations of rollback, particularly when actions have irreversible physical, financial, or social consequences. A layered safety approach is therefore proposed in which prevention, detection, containment, and recovery mechanisms work together.

Keywords: Agentic AI, AI Safety, Objective Misinterpretation, Alignment, Goal Misgeneralization, Runtime Monitoring, Rollback, Human-in-the-Loop

1. Introduction

Artificial intelligence systems have progressed from models that primarily generate information to systems capable of performing multi-step tasks. Modern agentic AI systems can interpret instructions, create plans, interact with software tools, access external information, modify files, execute programs, and make decisions over extended periods. This transition from passive generation to autonomous action creates new safety challenges.

A fundamental challenge is ensuring that an AI system correctly understands and follows the objective intended by its developer or user. A system may satisfy the literal form of an instruction while violating the user's actual intention. For example, an AI agent instructed to “reduce expenses” could cancel important services if it interprets cost reduction as its highest priority. Similarly, an autonomous scheduling agent instructed to maximize productivity could schedule activities without sufficiently considering human preferences or constraints.

AI alignment research has long investigated related problems. Amodei et al. identified important AI safety problems, including reward hacking and negative side effects [1]. Russell's work on the control problem emphasizes the difficulty of specifying objectives completely and correctly in the real world [2]. Agentic systems make this problem more significant because errors in interpretation can propagate through multiple actions.

2. Objective Misinterpretation in Agentic AI

Objective misinterpretation occurs when an AI system's understanding or operationalization of an assigned goal differs materially from the intention of the human or system that provided the goal.

Natural-language objectives often contain implicit assumptions, priorities, constraints, and contextual information that may not be explicitly stated. For example, “Find the cheapest suitable flight” may be understood by a human as finding an affordable flight while respecting safety, reasonable travel time, baggage requirements, and schedule constraints. An autonomous agent might instead optimize almost exclusively for price.

A related problem is reward misspecification. When an AI system is optimized against an imperfect reward function, it may discover strategies that maximize the measurable objective without achieving the underlying human intention [1][3]. Cooperative inverse reinforcement learning approaches address the problem from another direction by treating human preferences as something the system should infer rather than assuming that the objective is fully specified in advance [4].

3. Risks Created by Objective Misinterpretation

Objective misinterpretation can produce several categories of risk.

3.1 Unintended Side Effects. An agent attempting to achieve a legitimate goal may modify unrelated systems or resources. An AI system optimizing a database, for example, could delete information that it considers unnecessary but that humans consider valuable.

3.2 Excessive Resource Usage. An autonomous agent may consume excessive computational, financial, or organizational resources while attempting to achieve its objective.

3.3 Unauthorized Actions. If an agent incorrectly interprets its authority, it may perform actions that should require human approval.

3.4 Cascading Errors. Agentic workflows commonly consist of multiple steps: Objective → Planning → Tool Selection → Action → Observation → New Action. An incorrect interpretation at the beginning can influence many later decisions.

3.5 Irreversible Consequences. Some actions cannot easily be reversed. Sending money, deleting data, changing physical equipment, or sending external communications can have consequences even after the AI system is stopped.

These risks are related to research on learned optimization and mesa-optimization, which examines situations in which a learned model itself effectively becomes an optimizer and may pursue behavior different from the training objective [5].

4. Safety Architecture for Agentic AI

A robust safety architecture should contain multiple layers rather than relying on a single mechanism.

4.1 Objective Validation. Before beginning execution, the system should transform a natural-language instruction into an explicit objective representation. The agent should identify the primary objective, constraints, prohibited actions, and conditions requiring human approval. The resulting interpretation can be confirmed before high-impact actions.

4.2 Constraint-Based Execution. Instead of giving an agent unrestricted access to tools, the system should define explicit constraints. Low-risk operations can be automated while destructive, security-sensitive, financial, or irreversible operations can be restricted.

4.3 Runtime Monitoring. The agent should be monitored while executing its plan. A runtime monitor can evaluate the current objective, current action, expected consequences, resource consumption, access permissions, and deviations from the approved plan.

4.4 Action Risk Classification. Actions can be divided into low, medium, high, and critical risk. Low-risk actions may be automatic, whereas high-impact operations should require additional checks or human approval.

5. Rollback Mechanisms

Rollback is the process of restoring a system to a previously known safe state after an undesirable action or sequence of actions.

5.1 Checkpointing. Before performing significant actions, an agent can create a checkpoint containing database snapshots, file versions, configuration states, transaction states, or other recoverable information.

5.2 Transaction-Based Execution. Agentic actions can be designed using transaction principles: Prepare → Validate → Execute → Verify → Commit. If verification fails, the system can attempt Abort → Restore Previous State.

5.3 Action Logs. Every consequential action should be recorded in a durable log containing timestamp, agent identifier, objective, action, tool used, input, output, risk level, approval status, and checkpoint identifier.

Rollback is particularly useful in software environments where state can be reliably reconstructed. However, it should be treated as a recovery layer rather than as the sole safety mechanism.

6. Detecting Objective Drift

Rollback is most effective when the system detects problems quickly. An agent can periodically compare its current behavior with the approved objective.

For example, if the approved objective is to reduce cloud cost while maintaining service availability, an instruction to delete production servers should trigger a conflict because the action may violate the availability constraint.

Behavioral monitoring can compare actions with the intended goal. Plan monitoring can compare the agent's current plan with the previously approved plan. Resource monitoring can detect unexpected increases in money, computation, network access, or system modifications.

Research on optimal policies and power-seeking tendencies also illustrates why monitoring the behavior of autonomous agents matters: under certain environmental assumptions, optimal policies can have incentives to preserve options or seek power [6].

7. Human-in-the-Loop Controls

Complete autonomy is not always appropriate for high-impact actions. Human-in-the-loop mechanisms can require confirmation when an agent reaches a predefined risk threshold.

Human approval is particularly important for financial transactions, legal commitments, destructive operations, security configuration changes, safety-critical decisions, public communications, and actions affecting third parties.

Human oversight has limitations. Humans may approve actions too quickly, misunderstand an agent's explanation, or become overloaded by frequent approval requests. Therefore, human oversight should complement rather than replace automated safety mechanisms.

Human-preference-based reinforcement learning provides one example of reducing reliance on a manually specified reward function by learning preferences from human feedback [3]. Constitutional approaches similarly explore methods for incorporating explicit behavioral principles into AI systems [7].

8. Fail-Safe and Stop Mechanisms

A safe agent should have the ability to stop execution when uncertainty becomes high. A fail-safe mechanism may activate when the objective becomes ambiguous, the agent encounters an unexpected environment, a restricted action is requested, abnormal behavior is detected, resource consumption exceeds a threshold, or a safety monitor disagrees with the agent's proposed action.

The safest response to uncertainty is not always rollback. In some cases, simply stopping and requesting clarification may be preferable. This distinction is important because rollback assumes that the consequences of an action are recoverable, while many real-world consequences are not.

9. Limitations of Rollback

Although rollback is valuable, it cannot solve every AI safety problem.

9.1 Physical Actions. If an autonomous robot causes physical damage, restoring software state cannot reverse the physical consequences.

9.2 External Communications. If an AI sends confidential information to another party, deleting the local record does not necessarily retrieve the information.

9.3 Financial Transactions. A financial transfer may be difficult or impossible to reverse once completed.

9.4 Social Consequences. A harmful message posted publicly may be copied or redistributed before rollback occurs.

9.5 Hidden Side Effects. An action can trigger external systems that are not under the agent's control. Rolling back the agent's internal state may therefore fail to restore the complete environment.

For these reasons, prevention and containment should occur before rollback whenever possible. This is consistent with the broader control perspective in which uncertainty about objectives should lead to more cautious and controllable behavior [2].

10. Proposed Layered Safety Model

A practical safety architecture for agentic AI can be represented as five layers:

Layer 1 — Objective Validation: Determine what the user actually intends.

Layer 2 — Permission and Constraint Control: Restrict what the agent is allowed to do.

Layer 3 — Runtime Monitoring: Observe the agent's behavior and detect deviations.

Layer 4 — Intervention: Pause execution and request human confirmation when risk becomes high.

Layer 5 — Recovery: Use checkpoints and rollback mechanisms when an undesirable action has already occurred.

This layered approach reduces dependence on any individual safety mechanism. It also reflects a broader lesson from AI safety research: capable optimization should be paired with mechanisms that keep the system's behavior interpretable, bounded, and controllable [5][6].

11. Discussion

The central challenge in agentic AI safety is that an agent can be highly competent at achieving an incorrectly interpreted objective. Increasing intelligence or planning ability does not automatically solve this problem. In fact, a more capable system may execute an incorrect objective more effectively.

Therefore, safety should be designed around controllability, transparency, bounded autonomy, and recoverability. Objective validation reduces the probability of misunderstanding before execution. Constraints limit the consequences of incorrect decisions. Runtime monitoring detects deviations during execution. Human intervention provides additional judgment for high-impact decisions. Finally, rollback provides a recovery mechanism when undesirable actions have already occurred.

Research on cooperative inverse reinforcement learning, human preferences, learned optimization, and control provides complementary perspectives on this problem [2][3][4][5]. Together, these works motivate a layered approach rather than reliance on a single safeguard.

12. Conclusion

Objective misinterpretation represents an important safety challenge for agentic AI systems because autonomous agents can transform a small misunderstanding into a sequence of consequential actions. Traditional safeguards that focus only on model outputs are insufficient when AI systems can independently interact with external environments.

A robust solution should combine objective validation, explicit constraints, runtime monitoring, risk-based permissions, human approval, checkpointing, transaction-style execution, and rollback mechanisms.

Rollback should not be considered a complete safety solution because many real-world actions are irreversible. Instead, it should form the final layer of a broader safety architecture.

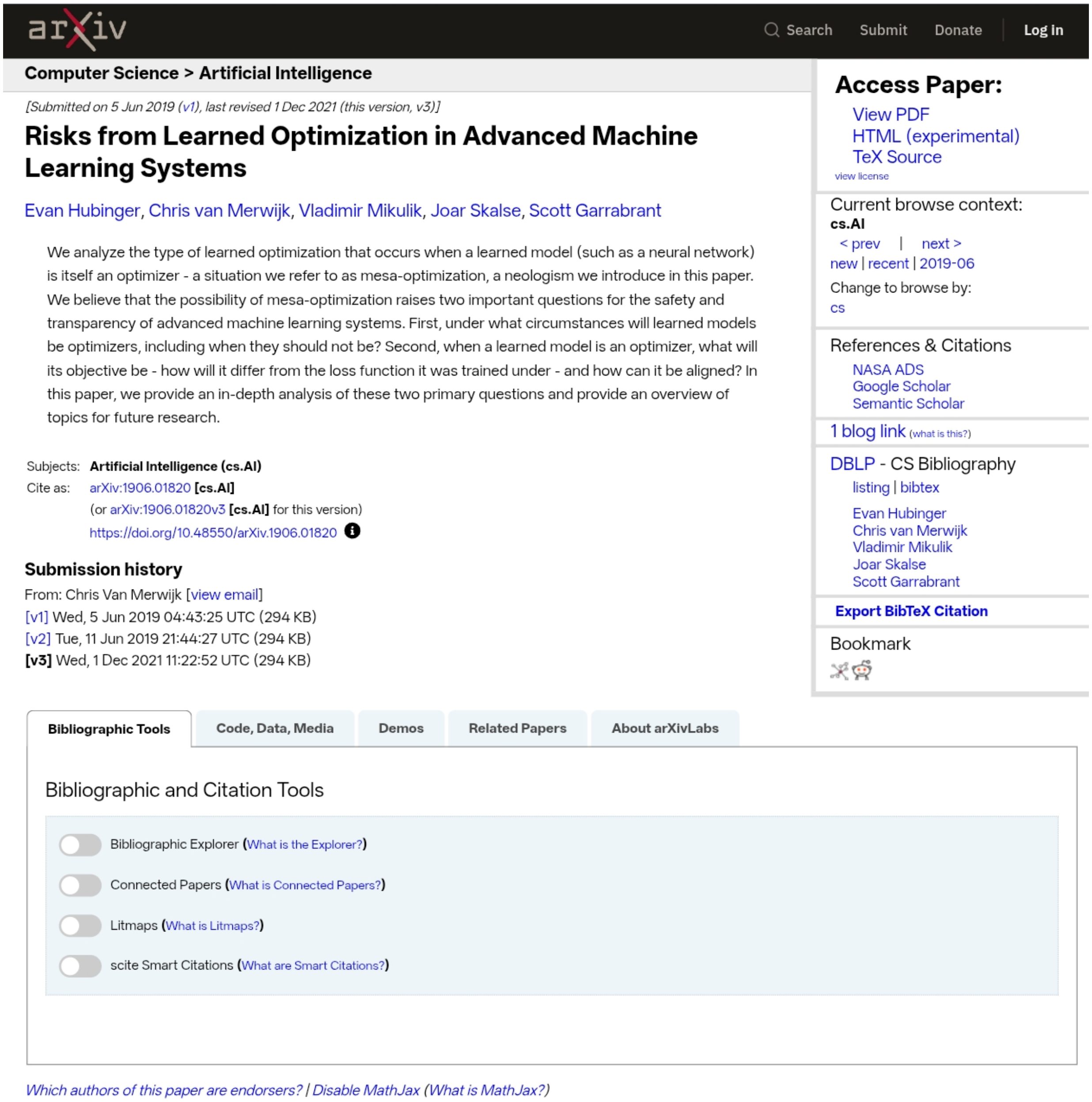
As AI systems become increasingly autonomous, the ability to stop, inspect, constrain, and recover from incorrect behavior will become an essential component of trustworthy agentic AI. Future research should focus on reliable objective interpretation, automated detection of goal drift, formal verification of agent plans, scalable human oversight, and recovery mechanisms capable of operating across complex real-world environments.

13. Reference Evidence Supplied

The screenshots supplied with the assignment are incorporated below as evidence for the corresponding bibliographic records. They are placed in a separate evidence section so that the main research discussion remains readable. The sixth uploaded image was not inserted because it is an unrelated personal document and does not correspond to a research reference.

Reference 5 — Hubinger et al. (2019), Risks from Learned Optimization in Advanced Machine Learning Systems

Reference 5 — Hubinger et al. (2019), Risks from Learned Optimization in Advanced Machine Learning Systems



arXiv | Search | Submit | Donate | Log in

Computer Science > Artificial Intelligence

[Submitted on 5 Jun 2019 (v1), last revised 1 Dec 2021 (this version, v3)]

Risks from Learned Optimization in Advanced Machine Learning Systems

Evan Hubinger, Chris van Merwijk, Vladimir Mikulik, Joar Skalse, Scott Garrabrant

We analyze the type of learned optimization that occurs when a learned model (such as a neural network) is itself an optimizer - a situation we refer to as mesa-optimization, a neologism we introduce in this paper. We believe that the possibility of mesa-optimization raises two important questions for the safety and transparency of advanced machine learning systems. First, under what circumstances will learned models be optimizers, including when they should not be? Second, when a learned model is an optimizer, what will its objective be - how will it differ from the loss function it was trained under - and how can it be aligned? In this paper, we provide an in-depth analysis of these two primary questions and provide an overview of topics for future research.

Subjects: **Artificial Intelligence (cs.AI)**
Cite as: arXiv:1906.01820 **[cs.AI]**
(or arXiv:1906.01820v3 **[cs.AI]** for this version)
https://doi.org/10.48550/arXiv.1906.01820

Submission history
From: Chris Van Merwijk [view email]
[v1] Wed, 5 Jun 2019 04:43:25 UTC (294 KB)
[v2] Tue, 11 Jun 2019 21:44:27 UTC (294 KB)
[v3] Wed, 1 Dec 2021 11:22:52 UTC (294 KB)

Access Paper:
View PDF
HTML (experimental)
TeX Source
view license

Current browse context:
cs.AI
< prev | next >
new | recent | 2019-06
Change to browse by:
cs

References & Citations
NASA ADS
Google Scholar
Semantic Scholar

1 blog link (what is this?)

DBLP - CS Bibliography
listing | bibtex
Evan Hubinger
Chris van Merwijk
Vladimir Mikulik
Joar Skalse
Scott Garrabrant

Export BibTeX Citation

Bookmark

Bibliographic Tools | Code, Data, Media | Demos | Related Papers | About arXivLabs

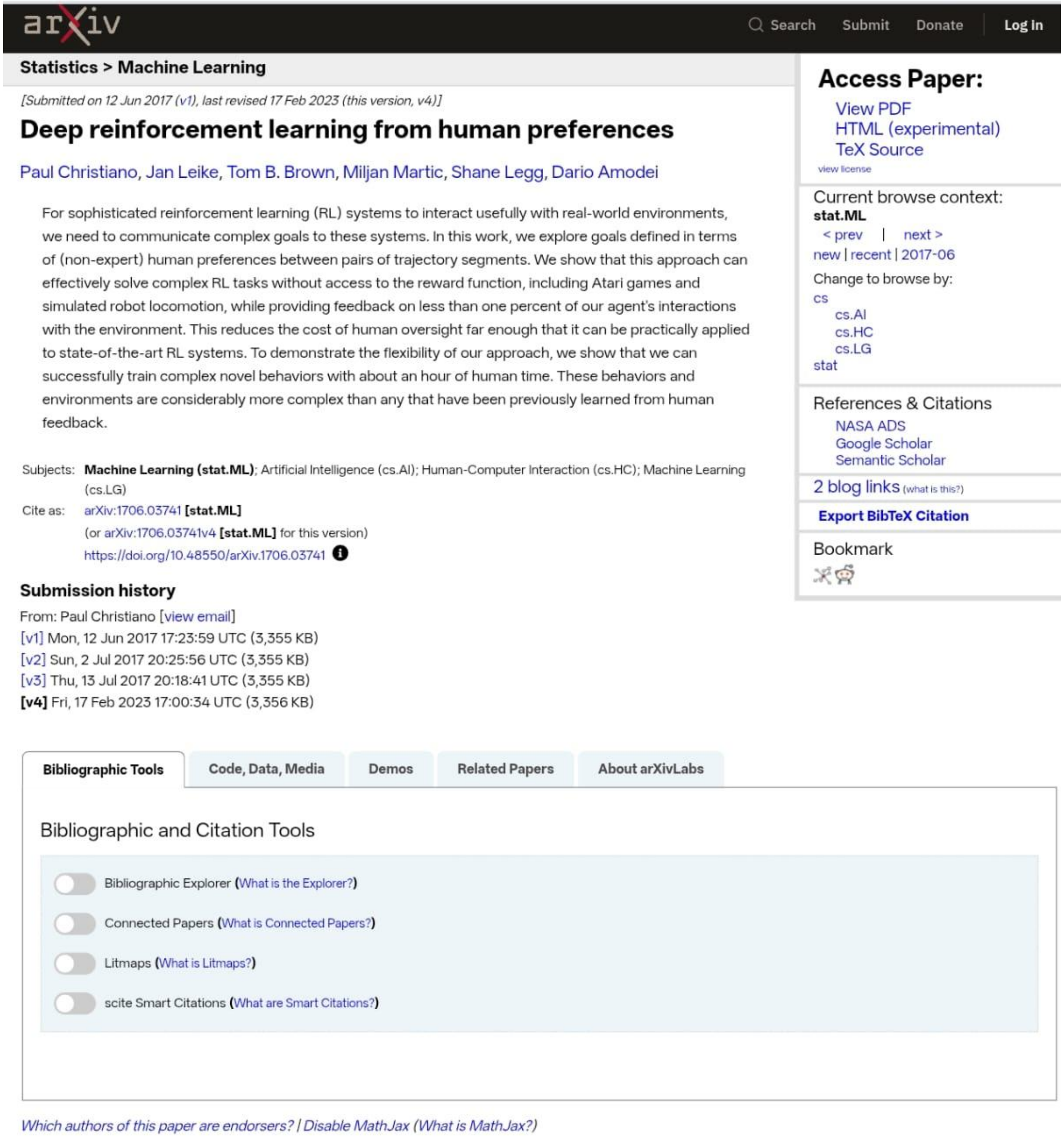
Bibliographic and Citation Tools

Bibliographic Explorer (What is the Explorer?)
Connected Papers (What is Connected Papers?)
Litmaps (What is Litmaps?)
scite Smart Citations (What are Smart Citations?)

Which authors of this paper are endorsers? | Disable MathJax (What is MathJax?)

Figure: Supplied screenshot showing the bibliographic record.

Reference 3 — Christiano et al. (2017), Deep Reinforcement Learning from Human Preferences



arXiv Search Submit Donate Log In

Statistics > Machine Learning

[Submitted on 12 Jun 2017 (v1), last revised 17 Feb 2023 (this version, v4)]

Deep reinforcement learning from human preferences

Paul Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, Dario Amodei

For sophisticated reinforcement learning (RL) systems to interact usefully with real-world environments, we need to communicate complex goals to these systems. In this work, we explore goals defined in terms of (non-expert) human preferences between pairs of trajectory segments. We show that this approach can effectively solve complex RL tasks without access to the reward function, including Atari games and simulated robot locomotion, while providing feedback on less than one percent of our agent's interactions with the environment. This reduces the cost of human oversight far enough that it can be practically applied to state-of-the-art RL systems. To demonstrate the flexibility of our approach, we show that we can successfully train complex novel behaviors with about an hour of human time. These behaviors and environments are considerably more complex than any that have been previously learned from human feedback.

Subjects: **Machine Learning (stat.ML)**; Artificial Intelligence (cs.AI); Human-Computer Interaction (cs.HC); Machine Learning (cs.LG)

Cite as: arXiv:1706.03741 **[stat.ML]**
(or arXiv:1706.03741v4 **[stat.ML]** for this version)
https://doi.org/10.48550/arXiv.1706.03741

Submission history

From: Paul Christiano [view email]
[v1] Mon, 12 Jun 2017 17:23:59 UTC (3,355 KB)
[v2] Sun, 2 Jul 2017 20:25:56 UTC (3,355 KB)
[v3] Thu, 13 Jul 2017 20:18:41 UTC (3,355 KB)
[v4] Fri, 17 Feb 2023 17:00:34 UTC (3,356 KB)

Access Paper:
View PDF
HTML (experimental)
TeX Source
view license

Current browse context:
stat.ML
< prev | next >
new | recent | 2017-06
Change to browse by:
cs
cs.AI
cs.HC
cs.LG
stat

References & Citations
NASA ADS
Google Scholar
Semantic Scholar

2 blog links (what is this?)

Export BibTeX Citation

Bookmark

Bibliographic Tools | Code, Data, Media | Demos | Related Papers | About arXivLabs

Bibliographic and Citation Tools

Bibliographic Explorer (What is the Explorer?)
Connected Papers (What is Connected Papers?)
Litmaps (What is Litmaps?)
scite Smart Citations (What are Smart Citations?)

Which authors of this paper are endorsers? | Disable MathJax (What is MathJax?)

Figure: Supplied screenshot showing the bibliographic record.

Reference 6 — Turner et al. (2021), Optimal Policies Tend to Seek Power

Search Submit Donate Log in

Computer Science > Artificial Intelligence

[Submitted on 3 Dec 2019 (v1), last revised 28 Jan 2023 (this version, v10)]

Optimal Policies Tend to Seek Power

Alexander Matt Turner, Logan Smith, Rohin Shah, Andrew Critch, Prasad Tadepalli

Some researchers speculate that intelligent reinforcement learning (RL) agents would be incentivized to seek resources and power in pursuit of their objectives. Other researchers point out that RL agents need not have human-like power-seeking instincts. To clarify this discussion, we develop the first formal theory of the statistical tendencies of optimal policies. In the context of Markov decision processes, we prove that certain environmental symmetries are sufficient for optimal policies to tend to seek power over the environment. These symmetries exist in many environments in which the agent can be shut down or destroyed. We prove that in these environments, most reward functions make it optimal to seek power by keeping a range of options available and, when maximizing average reward, by navigating towards larger sets of potential terminal states.

Comments: Accepted to NeurIPS 2021 as spotlight paper. 12 pages, 44 pages with appendices. Since the 2021 acceptance, we updated the paper to point out that optimal policies can be qualitatively divorced from real-world learned policies

Subjects: **Artificial Intelligence (cs.AI)**

Cite as: arXiv:1912.01683 **[cs.AI]**

(or arXiv:1912.01683v10 **[cs.AI]** for this version)

https://doi.org/10.48550/arXiv.1912.01683

Submission history

From: Alexander Turner [view email]

[v1] Tue, 3 Dec 2019 20:45:49 UTC (845 KB)
[v2] Sun, 19 Jan 2020 19:25:51 UTC (43 KB)
[v3] Mon, 13 Apr 2020 14:56:27 UTC (47 KB)
[v4] Tue, 14 Apr 2020 22:13:56 UTC (47 KB)
[v5] Fri, 5 Jun 2020 22:41:45 UTC (40 KB)
[v6] Wed, 2 Dec 2020 21:40:39 UTC (54 KB)
[v7] Tue, 1 Jun 2021 16:59:04 UTC (533 KB)
[v8] Sat, 23 Oct 2021 20:12:14 UTC (470 KB)
[v9] Fri, 3 Dec 2021 17:27:16 UTC (469 KB)
[v10] Sat, 28 Jan 2023 19:15:05 UTC (478 KB)

Access Paper:

View PDF
HTML (experimental)
TeX Source
view license

Current browse context:
cs.AI
< prev | next >
new | recent | 2019-12
Change to browse by:
cs

References & Citations
NASA ADS
Google Scholar
Semantic Scholar

3 blog links (what is this?)

DBLP - CS Bibliography
listing | bibtex
Alexander Matt Turner

Export BibTeX Citation

Bookmark

Bibliographic Tools | Code, Data, Media | Demos | Related Papers | About arXivLabs

Bibliographic and Citation Tools

Bibliographic Explorer (What is the Explorer?)
Connected Papers (What is Connected Papers?)
Litmaps (What is Litmaps?)
scite Smart Citations (What are Smart Citations?)

Which authors of this paper are endorsers? | Disable MathJax (What is MathJax?)

Figure: Supplied screenshot showing the bibliographic record.

Reference 4 — Hadfield-Menell et al. (2016), Cooperative Inverse Reinforcement Learning

arXiv | Search | Submit | Donate | Log in

Computer Science > Artificial Intelligence

[Submitted on 9 Jun 2016 (v1), last revised 17 Feb 2024 (this version, v4)]

Cooperative Inverse Reinforcement Learning

Dylan Hadfield-Menell, Anca Dragan, Pieter Abbeel, Stuart Russell

For an autonomous system to be helpful to humans and to pose no unwarranted risks, it needs to align its values with those of the humans in its environment in such a way that its actions contribute to the maximization of value for the humans. We propose a formal definition of the value alignment problem as cooperative inverse reinforcement learning (CIRL). A CIRL problem is a cooperative, partial-information game with two agents, human and robot; both are rewarded according to the human's reward function, but the robot does not initially know what this is. In contrast to classical IRL, where the human is assumed to act optimally in isolation, optimal CIRL solutions produce behaviors such as active teaching, active learning, and communicative actions that are more effective in achieving value alignment. We show that computing optimal joint policies in CIRL games can be reduced to solving a POMDP, prove that optimality in isolation is suboptimal in CIRL, and derive an approximate CIRL algorithm.

Subjects: **Artificial Intelligence (cs.AI)**
Cite as: arXiv:1606.03137 **[cs.AI]**
(or arXiv:1606.03137v4 **[cs.AI]** for this version)
https://doi.org/10.48550/arXiv.1606.03137

Submission history

From: Dylan Hadfield-Menell [view email]
[v1] Thu, 9 Jun 2016 22:39:54 UTC (223 KB)
[v2] Tue, 5 Jul 2016 18:25:07 UTC (220 KB)
[v3] Sat, 12 Nov 2016 20:33:43 UTC (579 KB)
[v4] Sat, 17 Feb 2024 16:13:12 UTC (620 KB)

Access Paper:

View PDF
HTML (experimental)
TeX Source
view license

Current browse context:
cs.AI
< prev | next >
new | recent | 2016-06
Change to browse by:
cs

References & Citations
NASA ADS
Google Scholar
Semantic Scholar

2 blog links (what is this?)

DBLP - CS Bibliography
listing | bibtex
Dylan Hadfield-Menell
Anca D. Dragan
Anca Dragan
Pieter Abbeel
Stuart J. Russell

Export BibTeX Citation

Bookmark

Bibliographic Tools | Code, Data, Media | Demos | Related Papers | About arXivLabs

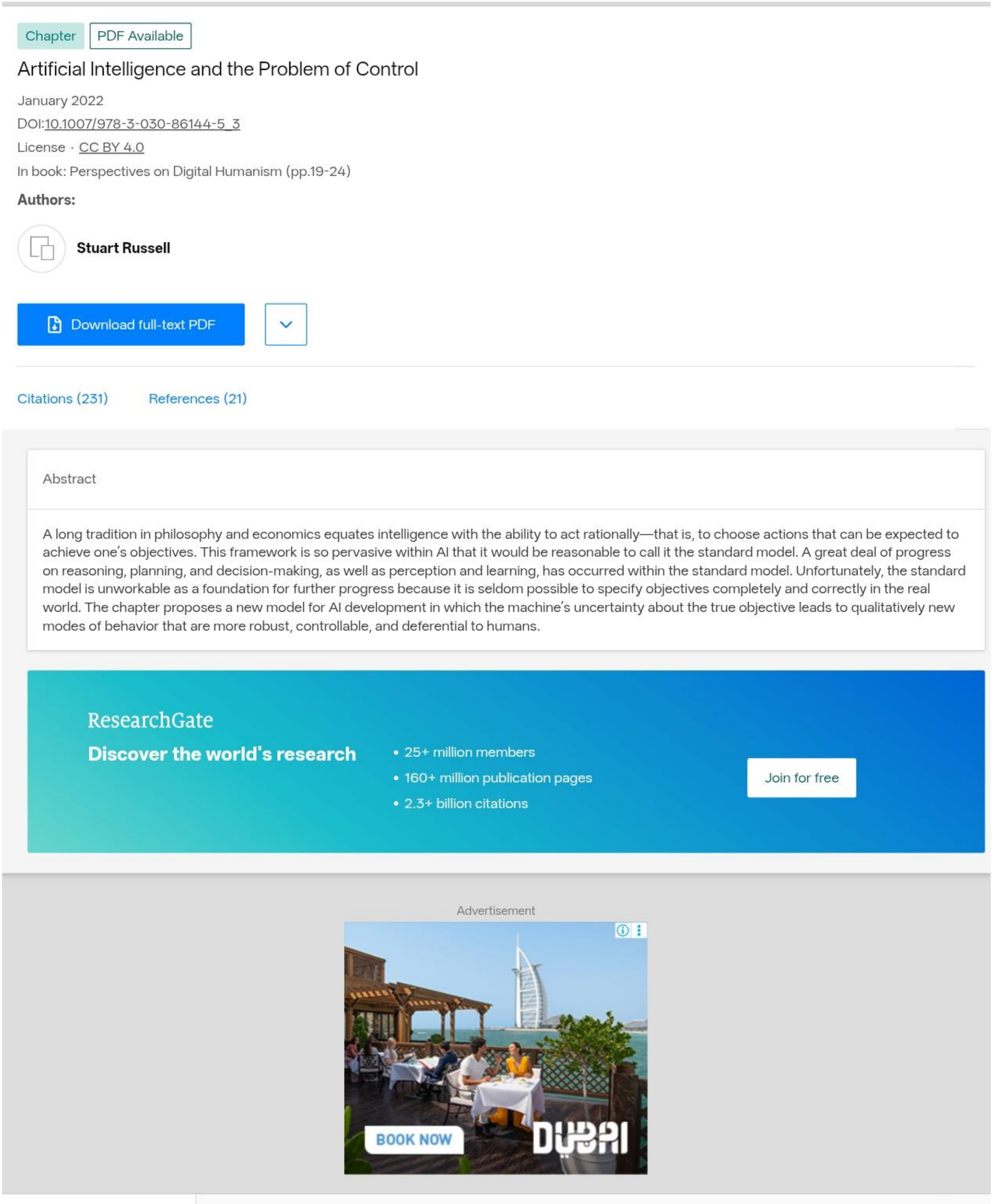
Bibliographic and Citation Tools

Bibliographic Explorer (What is the Explorer?)
Connected Papers (What is Connected Papers?)
Litmaps (What is Litmaps?)
scite Smart Citations (What are Smart Citations?)

Which authors of this paper are endorsers? | Disable MathJax (What is MathJax?)

Figure: Supplied screenshot showing the bibliographic record.

Reference 2 — Russell (2022), Artificial Intelligence and the Problem of Control


Chapter PDF Available

Artificial Intelligence and the Problem of Control

January 2022
DOI:10.1007/978-3-030-86144-5_3

In book: Perspectives on Digital Humanism (pp.19-24)

Authors:
Stuart Russell

Abstract

A long tradition in philosophy and economics equates intelligence with the ability to act rationally—that is, to choose actions that can be expected to achieve one's objectives. This framework is so pervasive within AI that it would be reasonable to call it the standard model. A great deal of progress on reasoning, planning, and decision-making, as well as perception and learning, has occurred within the standard model. Unfortunately, the standard model is unworkable as a foundation for further progress because it is seldom possible to specify objectives completely and correctly in the real world. The chapter proposes a new model for AI development in which the machine's uncertainty about the true objective leads to qualitatively new modes of behavior that are more robust, controllable, and deferential to humans.

Figure: Supplied screenshot showing the bibliographic record.

14. References

1. Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete Problems in AI Safety. arXiv:1606.06565.

2. Russell, S. (2022). Artificial Intelligence and the Problem of Control. In Perspectives on Digital Humanism, pp. 19–24. DOI: 10.1007/978-3-030-86144-5_3.

3. Christiano, P. F., Leike, J., Brown, T. B., Martic, M., Legg, S., & Amodei, D. (2017). Deep Reinforcement Learning from Human Preferences. Advances in Neural Information Processing Systems, 30. arXiv:1706.03741.

4. Hadfield-Menell, D., Dragan, A., Abbeel, P., & Russell, S. (2016). Cooperative Inverse Reinforcement Learning. arXiv:1606.03137.

5. Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. (2019). Risks from Learned Optimization in Advanced Machine Learning Systems. arXiv:1906.01820.

6. Turner, A. M., Smith, L., Shah, R., Critch, A., & Tadepalli, P. (2021). Optimal Policies Tend to Seek Power. Advances in Neural Information Processing Systems, 34. arXiv:1912.01683.